Neuromuscular conditions (Polyneuropathy)

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Learning Outcomes

By the end of this session, students should be able to:

- Describe the classification of neuromuscular conditions (disorders)
- Define what polyneuropathy is polyneuropathy and what is responsible for the signs and symptoms.
- Explain the pathophysiology of polyneuropathy
- Explain the overall management of polyneuropathy and the role of the physiotherapist
- Plan the management of a patient with polyneuropathy

Classification of neuromuscular disorders

α-motor neurone pathology

- **Poliomyelitis** *Spinal Muscular Atrophy, Motor Neurone Disease (MND/ALS)*

Nerve root

Spinal nerve

Plexus Brachial plexus

Peripheral nerve pathology

- Peripheral nerve lesion – Common peroneal nerve
- Neuropathy - *Guillain-Barré syndrome, CIDP, Charcot-Marie Tooth disease, Diabetic neuropathy*

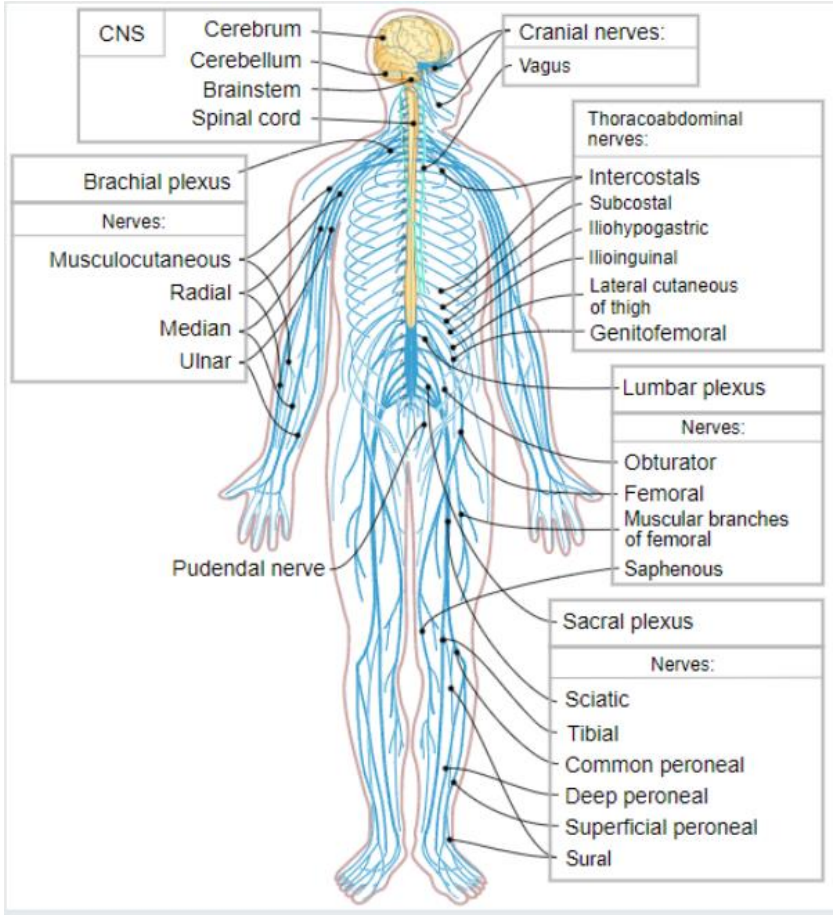
Neuromuscular junction disorders

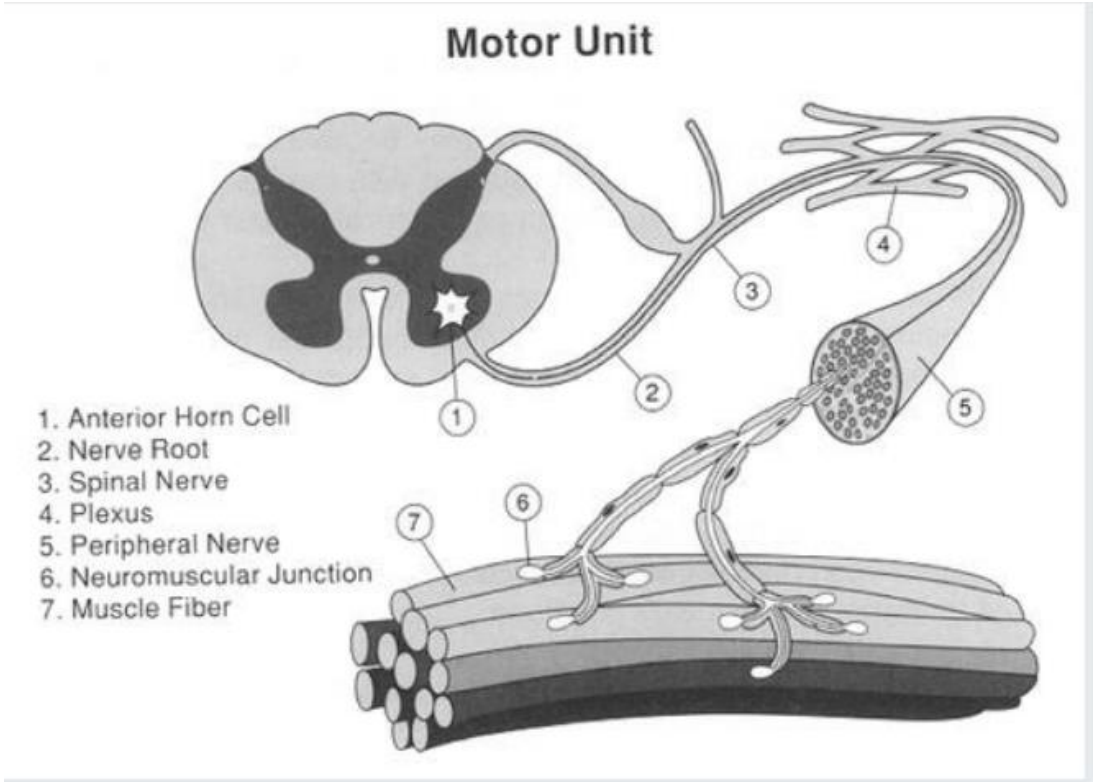
- *Myasthenia Gravis, Lambert Eaton Myasthenic Syndrome*

Muscle pathology or Myopathy

- *Inflammatory (myositis),*
- *Degenerative (dystrophies).*
- *Contractile (myotonias)*

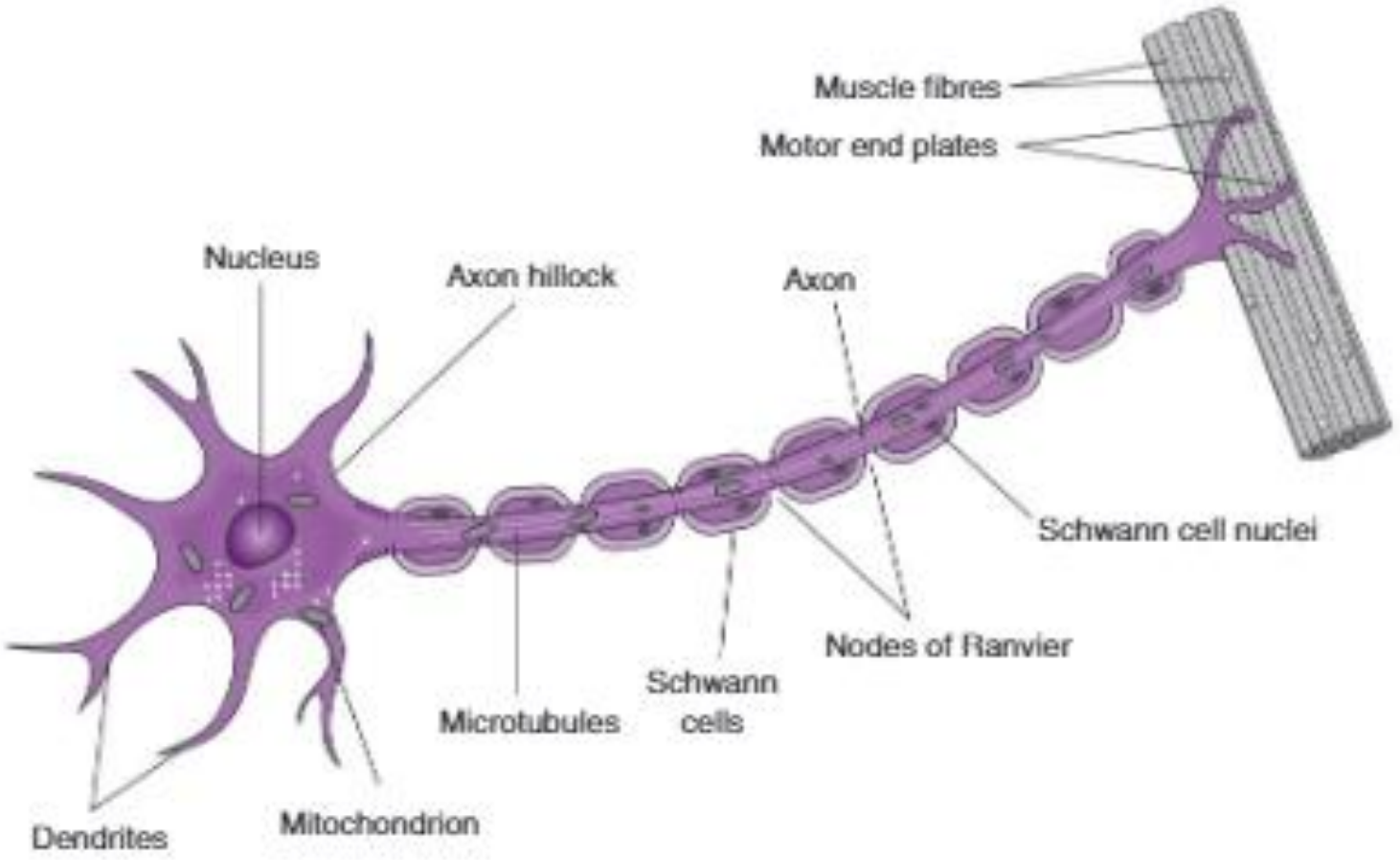
Neuromuscular conditions



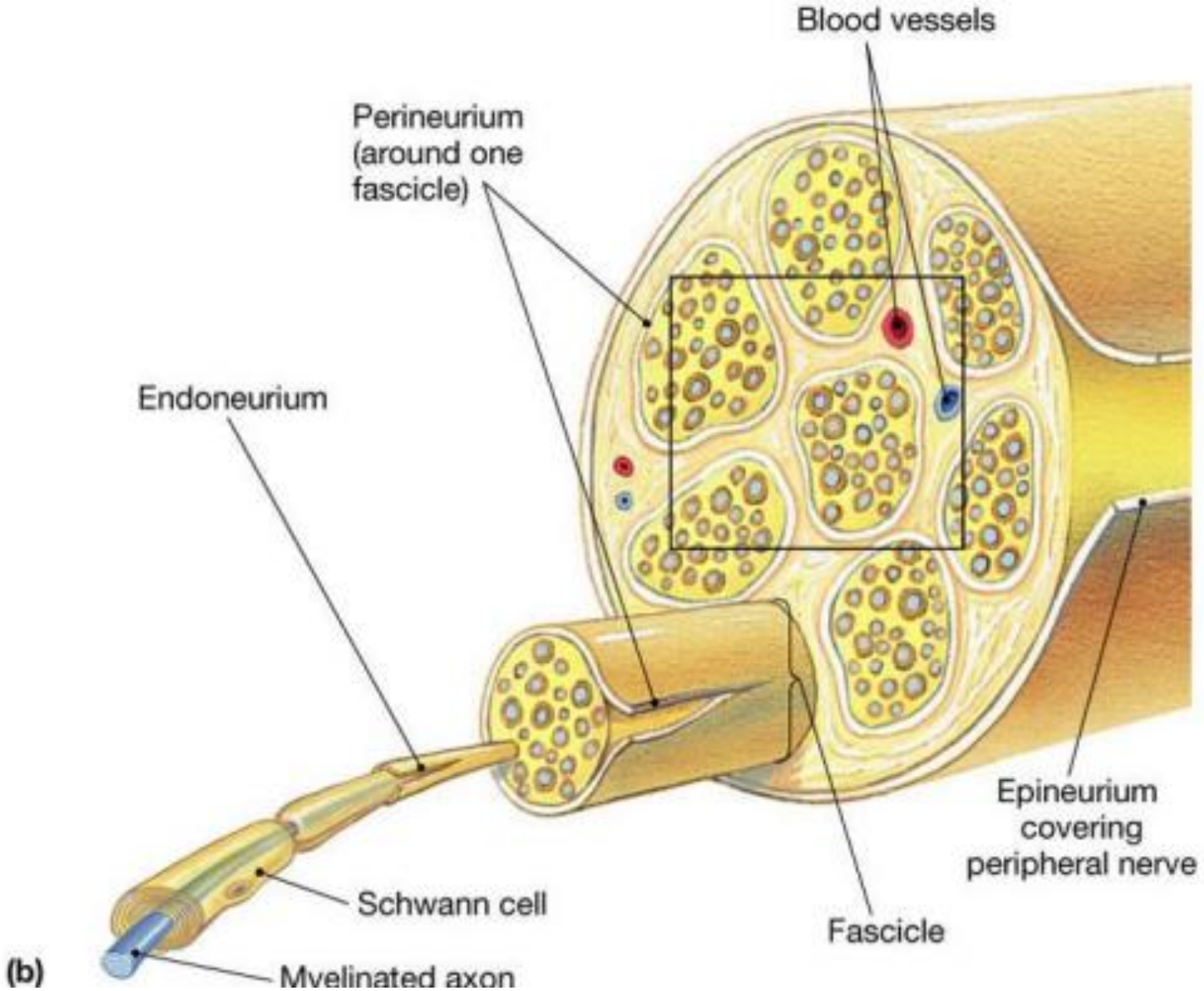


1. α-motor neurone,
2. Nerve roots
3. Spinal nerve
4. Plexus
5. Peripheral nerve
6. Neuromuscular junction
7. Muscle fibre

Structure of a motor neuron



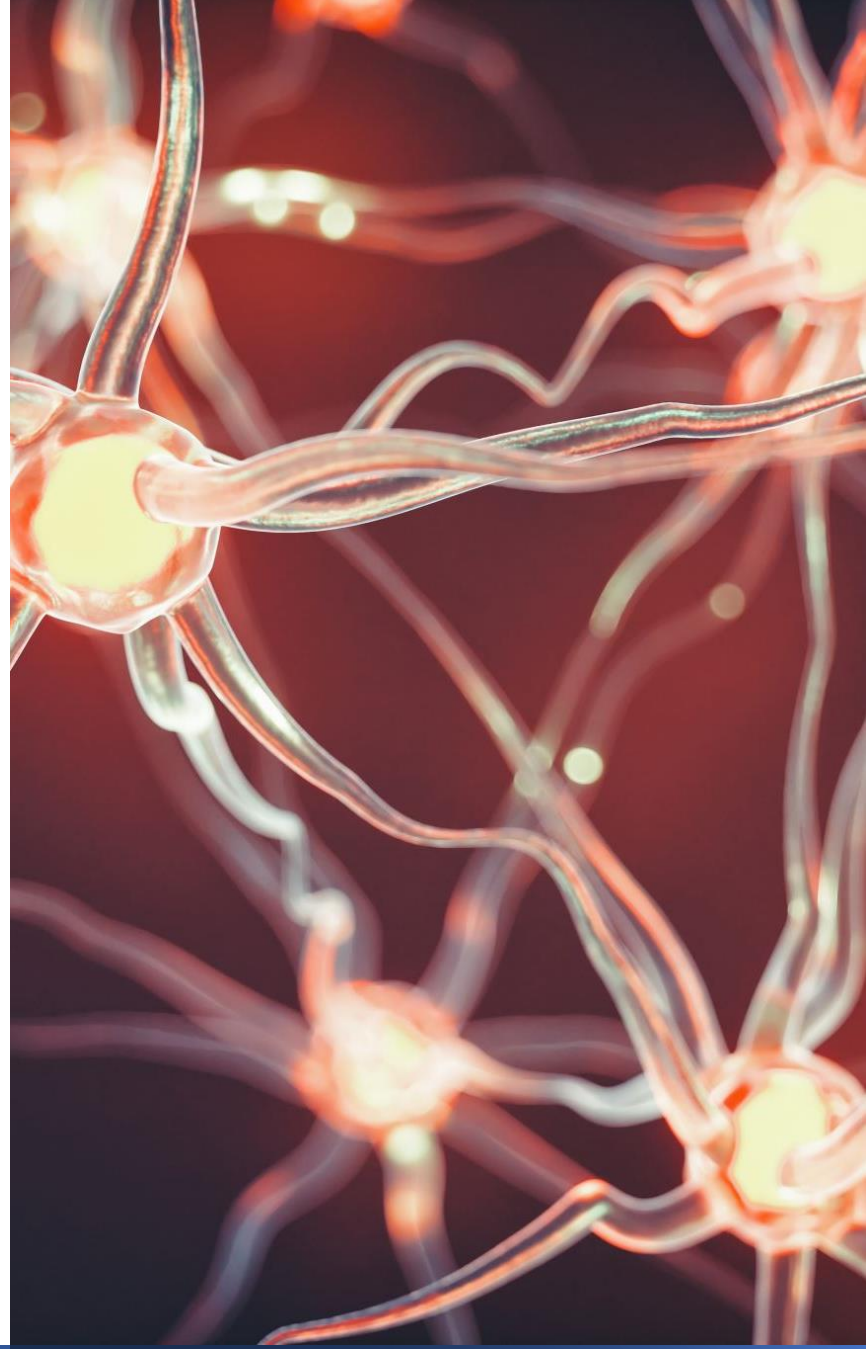
Structure of a peripheral nerve



Problems with peripheral nerves

- **<u>Neuropathy or polyneuropathies</u>**
  - Neuropathy (one nerve) or polyneuropathy (many nerves) are generalised disorders of the peripheral nerves
  - Varying causes
  - Some have acute onset and progression
  - Axonal degeneration, demyelination or vascular nerve damage

- **Peripheral Nerve Injuries**
  - Damage or injuries to peripheral nerves



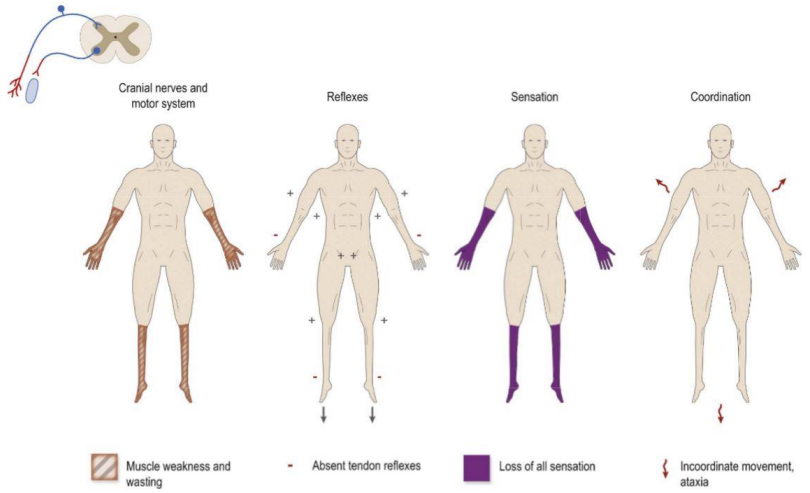
Pathology: – polyneuropathies

Polyneuropathy (disease)

- systemic disease (diabetes, HIV),
- drugs & toxins,
- inflammatory dysimmune,
- nutritional deficiency,
- hereditary neuropathy

Signs and Symptoms

- glove & stocking sensory loss or paraesthesia
- absent or hyporeflexia
- muscle weakness & atrophy
- autonomic neuropathy





Types of polyneuropathies

Types

- **Demyelinating neuropathy**
  - Acute demyelinating polyneuropathy e.g. Guillain Barre syndrome
  - Chronic demyelinating polyneuropathy e.g. Chronic inflammatory demyelinating polyneuropathy,
    - Chronic inflammatory demyelinating polyradiculopathy (CIDP)
    - Charcot-Marie –Tooth disease type 1 (CMT1)

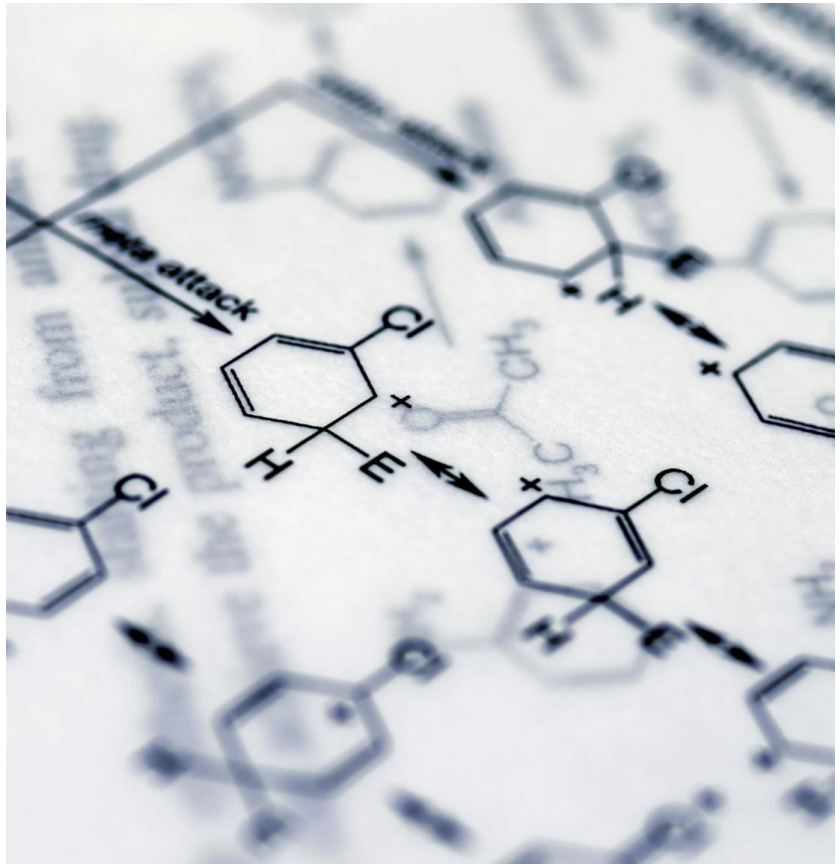
- **Axonal neuropathy**
  - Chronic axonal neuropathies are the most common
  - Causes vary from metabolic disorders , malignancies to toxicity from chemical agents
  - Degeneration of axons
  - Common examples are :
    - Diabetic neuropathy and Charcot-Maries Tooth disease type 2 (CMT 2)

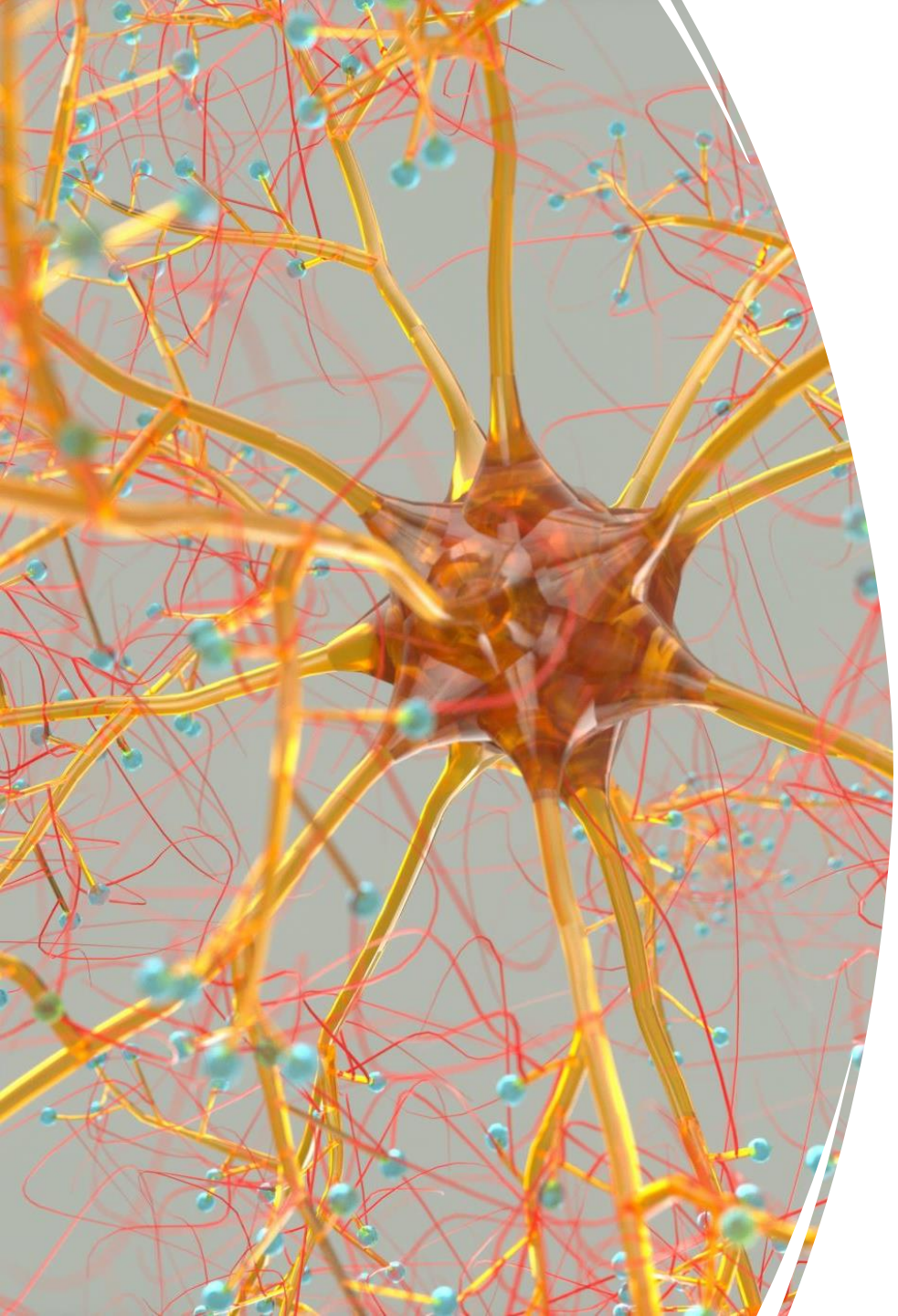


Pathology of peripheral neuropathy

- **Axonal degeneration**
  - Toxic, metabolic, nutritional and physical insults (cold or trauma)
  - Longest axons affected the most
  - Distal weakness, wasting or sensory loss
  - If neuron remain intact – regenerates, if not, no scope for regeneration.

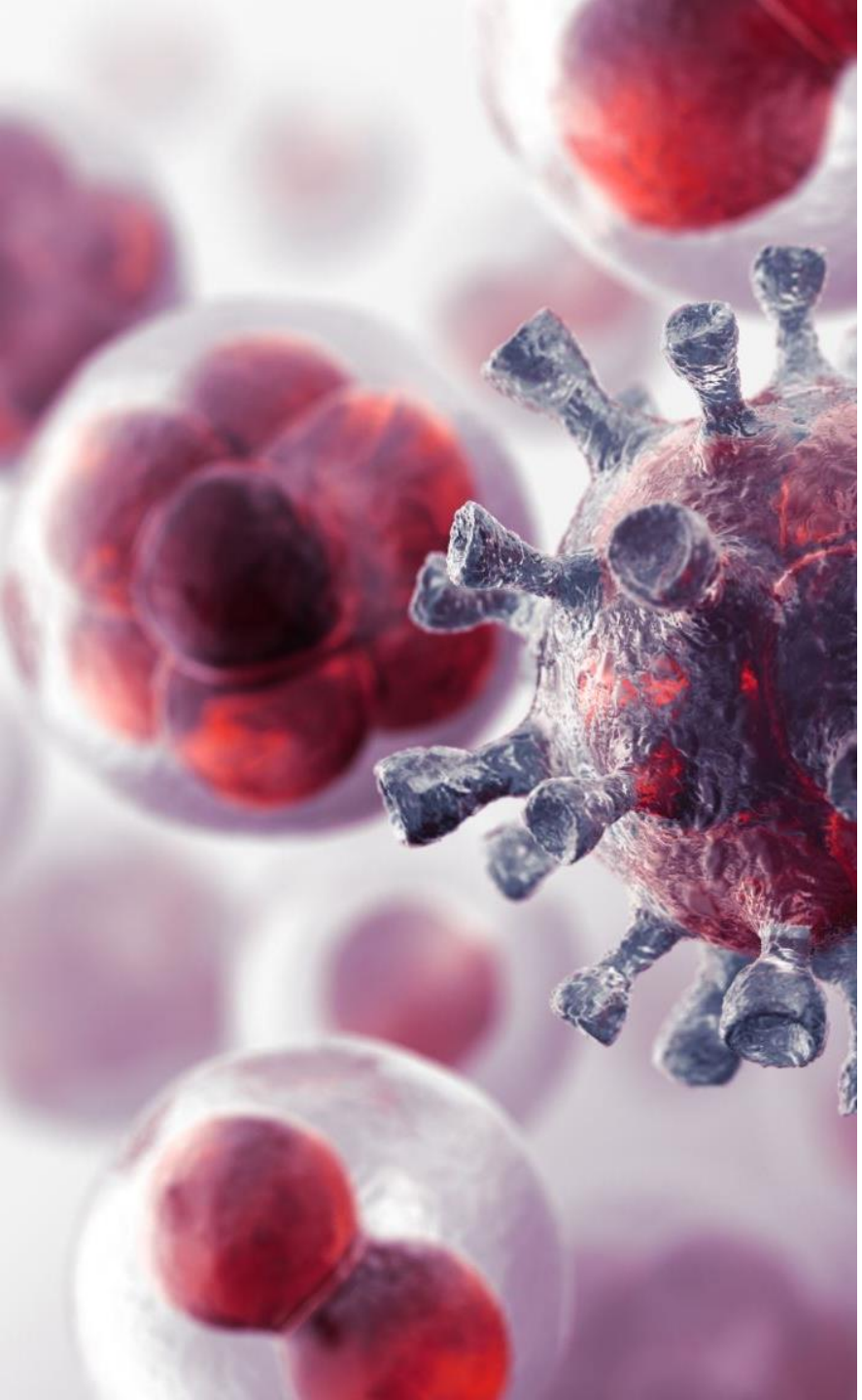
- **Demyelination**
  - Inflammatory diseases
  - Myelin is lost but the axon is preserved.
  - Demyelination prevents conduction and produces weakness but not wasting
  - Repair can be quick
  - May also involve axonal degeneration





Pathology of peripheral neuropathy 2

- **<u>Vascular nerve damage</u>**
  - Blood supply to peripheral nerves can be disrupted producing infarct of peripheral nerves
  - Leads to axonal degeneration in the axons distal to the site of the infarct
  - Occurs with vasculitis illness



Gullain Barre Syndrome (GBS)

- Auto-immune disorder - immune response is directed towards unknown antigens
- Incidence: 1.9 cases per 100000 population (European data)
- Can affect any age but most common in men and in older people

- Unknown causes but an infection precedes onset in a third of cases.
- The response leads to inflammatory process and destruction of myelin sheath in larger diameter motor and sensory neurons

- Rapidly progressive paralysis
- Sensory impairment
- Areflexia

Pathophysiology in GBS and clinical relevance

Inflammatory process is primary cause of symptoms in GBS of both demyelinating and axonal types but where axonal degeneration and bystander degeneration occurs in GBS this causes persistent problems

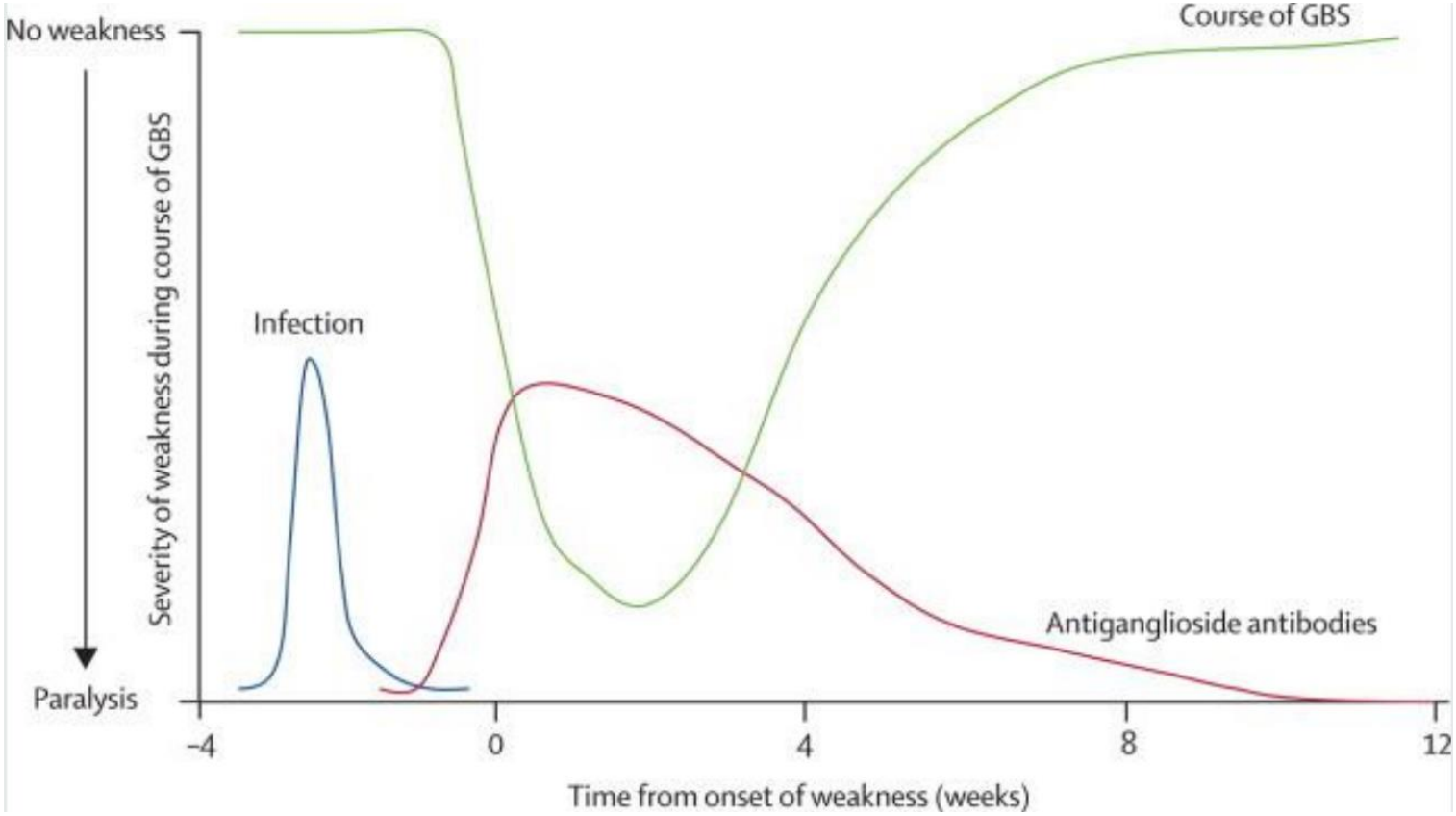
Remyelination after inflammation has resolved gives rise to functional recovery unless axonal degeneration has occurred

Collateral sprouting after axonal degeneration aids maintenance of function.

Extended periods of denervation may prevent reinnervation by regenerating axons and motor neuron death may occur resulting in permanent functional loss

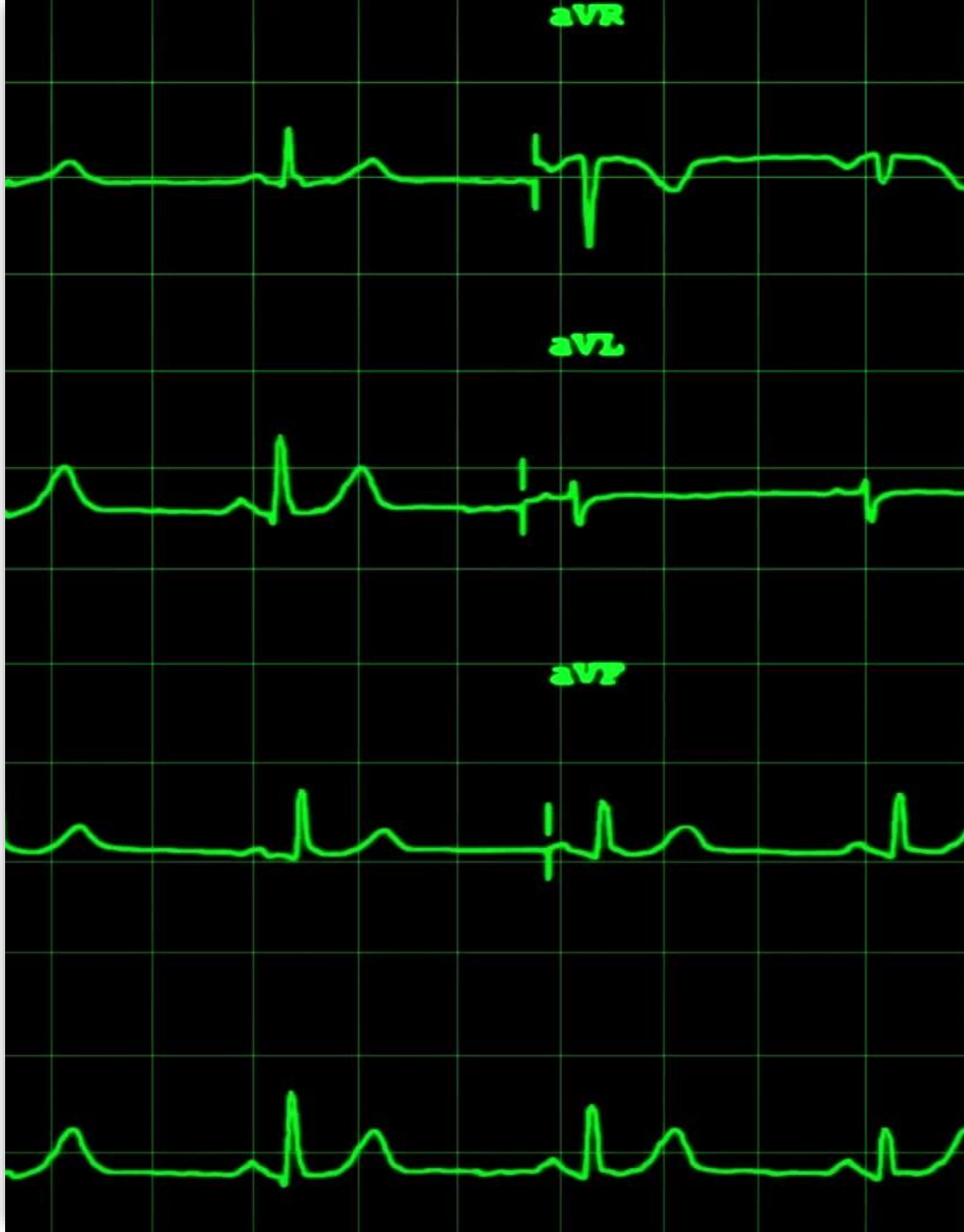
Expanded motor units as a result of collateral sprouting may decompensate many years after the initial insult leading to an increase in symptoms and disability

Acute neuromuscular conditions – Guillain Barre syndrome



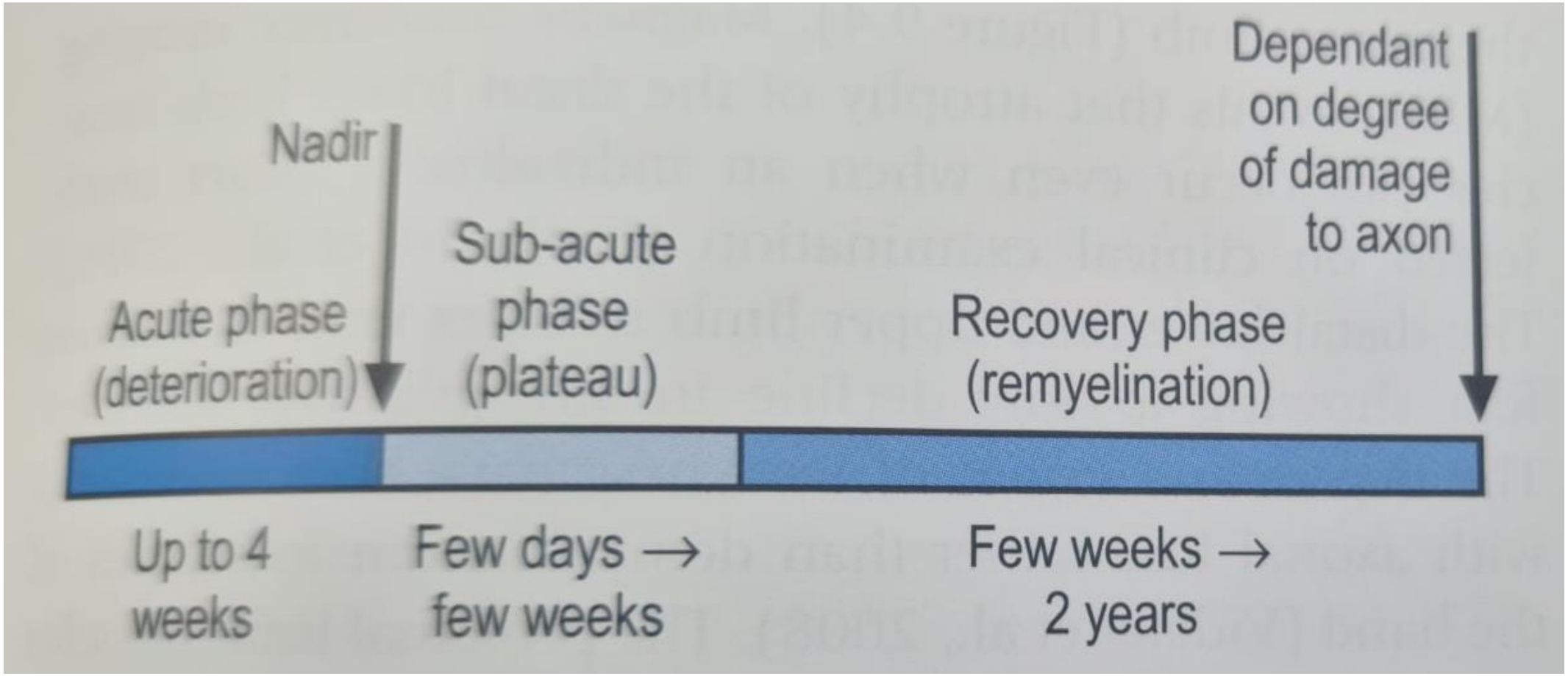
Phases and progression of GBS

- Blood tests demonstrates:
  - Elevated immunoglobulins in 25% of cases
  - Elevated proteins in CSF in 80% of cases
  - Nerve conduction studies show decreased conduction velocity
  - Partial conduction block
  - Decreased composite muscle action potential (CMAP) if there is axonal loss.



Phases and progression of GBS

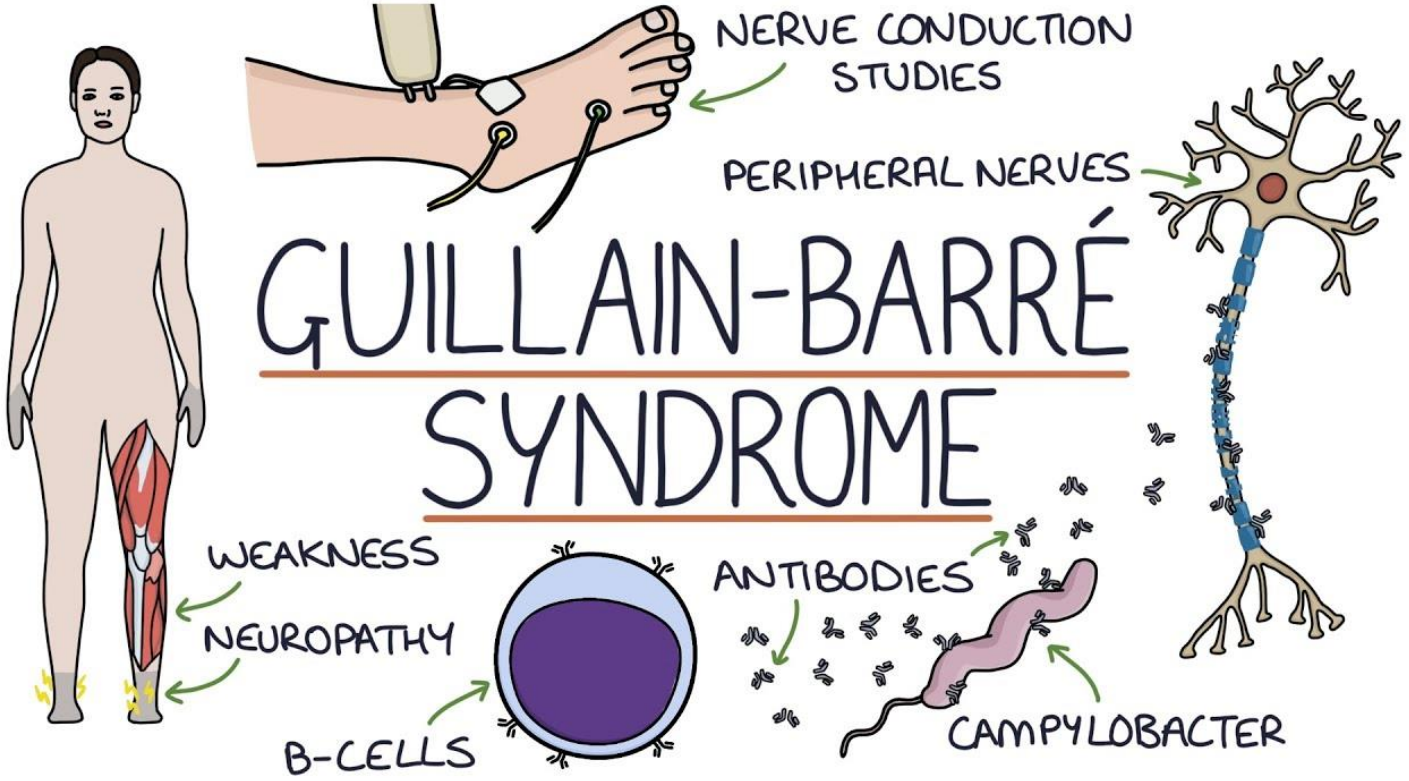
Relapses common in 16%
10 -20% a have residual disabling symptoms



Presentation of GBS

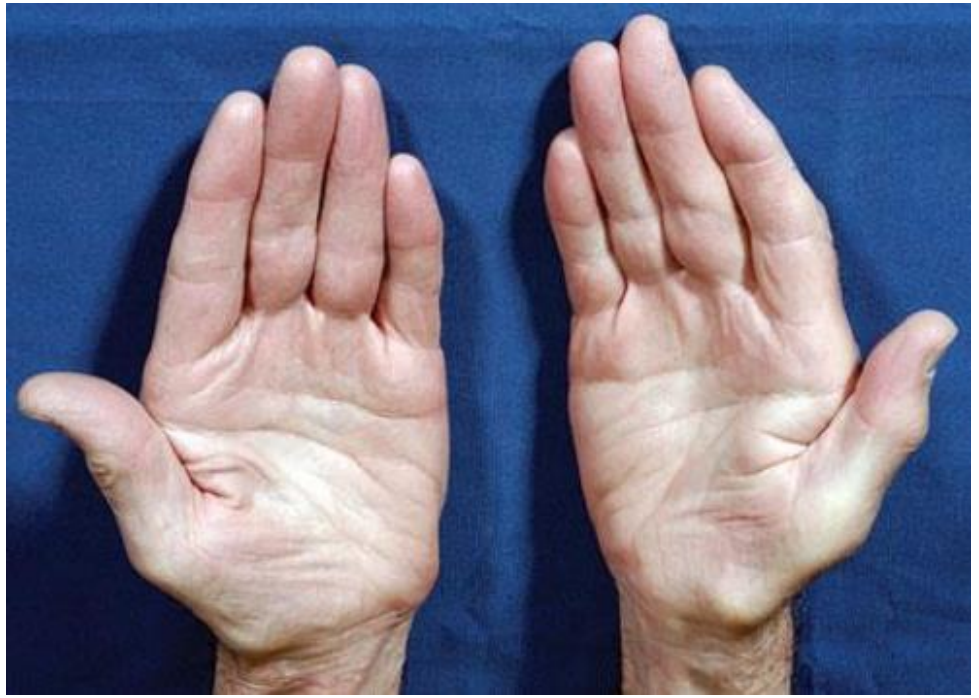
- Primary pathology: demyelination

- Presentation
  - Proximal or distal weakness and sensory loss
  - Ascends or descends from onset
  - Impact walking, standing or even sitting in severe cases
  - Facial and bulbar musculature (50%)
  - Respiratory function affected (1/3)
  - Autonomic involvement – Blood pressure and cardiac arrhythmias



Signs and symptoms (SS) of Lower motor neuron(LMN) pathology - motor signs

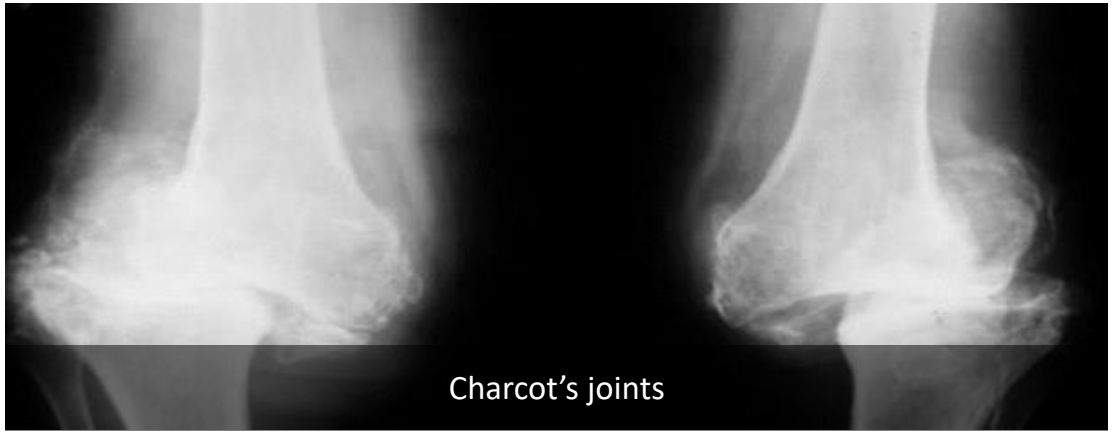




Signs and Symptoms of LMN pathology - negative neuropathic sensory symptoms



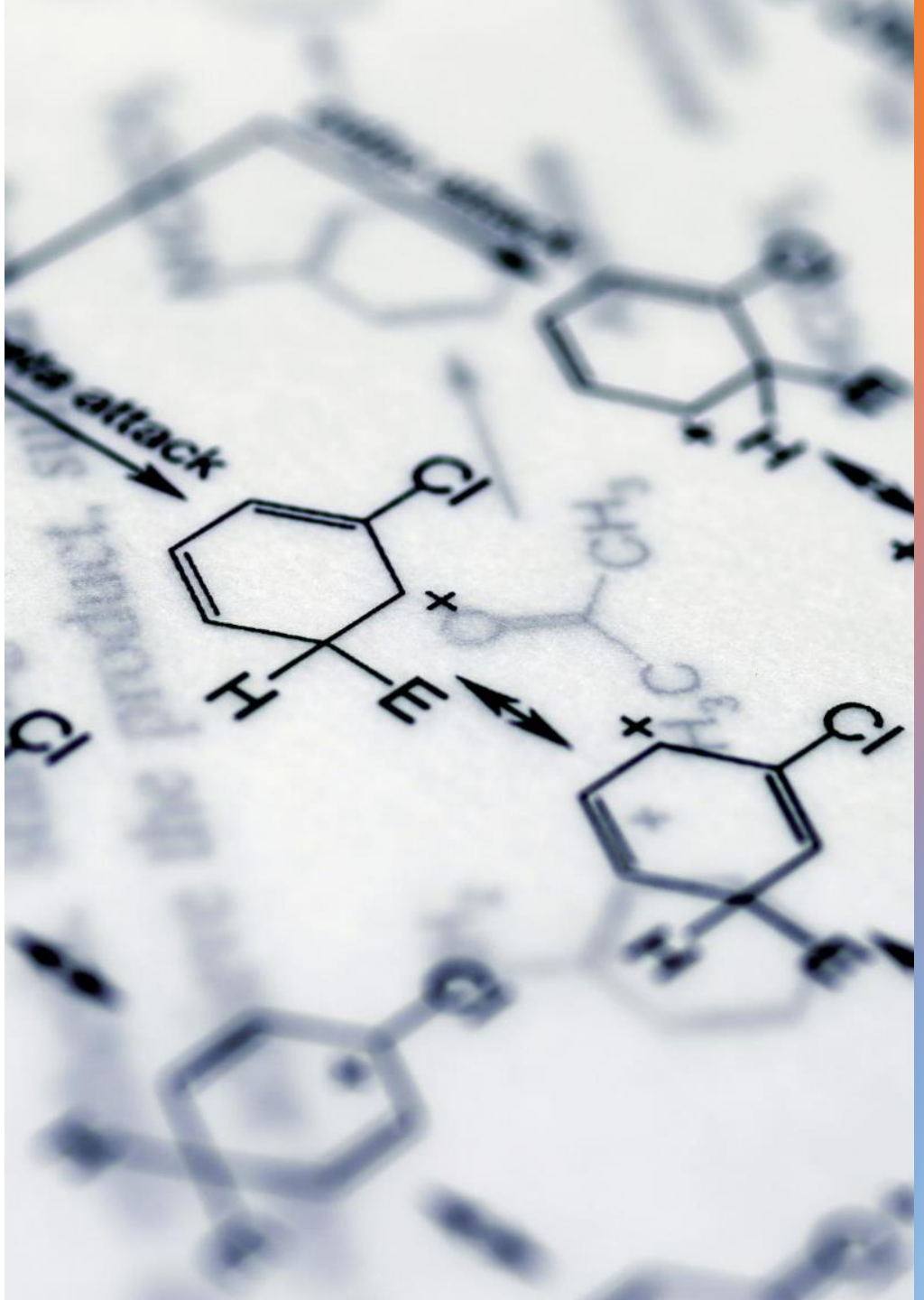
Digital damage – Hansen's disease (leprosy)



Charcot's joints

Signs & Symptoms of LMN pathology – Positive Neuropathic Sensory Symptoms (P-NSS)

- Paresthesia – Pins and needles, burning
- Hyperesthesia – general increase in sensitivity of any of the senses (not specific to touch)
- Hyperalgesia – increased pain from a noxious stimulus
- Allodynia – painful response to non-noxious stimuli



Signs & Symptoms of LMN pathology - Autonomic

- Autonomic
  - Trophic changes
  - vasomotor
  - sudomotor
  - pilomotor
- Pain / hyperesthesia
- Gastric, sexual, continence problems, Hypoglycaemia unawareness
- Cardiovascular problems



Autonomic Signs & Symptoms

Cardiovascular involvement

- Altered heart rate variability (HRV)
  - Reduced variability and response
- Orthostatic/postural hypotension
  - Lying to standing
- Tachycardia
- Altered HR response to exercise
  - Check response prior to prescription of exercise
  - Use Borg rating of perceived exertion rather than HR for monitoring intensity

Acute Management of GBS

Address any serious complications

- Reduced Vital capacity
- Airway protection
- Critical care support

Pharmacotherapy

- IV immunoglobulins
- Plasma exchange or plasmapheresis

Physiotherapy

- Respiratory interventions e.g. sputum clearance, maintaenance of lung volumes, breathing exercises
- Prevention of secondary complications – early mobilisation, splinting, positioning, stretches
- Communication to allay anxiety and confusion about prognosis



General management for polyneuropathy

- Exercise to improve muscle strength
- Exercise to improve aerobic fitness
- Functional re-education
- Balance training
- Orthotics management
- Fatigue management

Reference

- Stokes M & Stack E. Polyneuropathy. Chapter in Physical management for Neurological Conditions. Churchill Livingstone. Elservier

